

Ebike Speed Controller System

INTRODUCTION TO ENGINEERING

PROJECT REPORT

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MAY 2025



BONAFIDE CERTIFICATE

Certified that this project “**Ebike Speed Controller System**” is the bonafide work of **N Nithish Kumar(24UEEC0183), M Yeshwanth (24UEEC0336), P Supraja (24UEEC0506), Poorna Chandu (24UEEC0181), Koushik(24UEEC0021), Ganesh(24UEEC0019)** who carried out this project work under my supervision.

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The Examination is held on _____

EXAMINER 1

EXAMINER 2

ABSTRACT

E-bike Speed Controller System

Electric bicycles (E-bikes) have transformed personal transportation, providing an efficient and environmentally friendly alternative to conventional vehicles. With the growing adoption of E-bikes, speed regulation has become a crucial factor for ensuring both safety and efficiency. Traditional speed control methods often suffer from imprecision and a lack of adaptability, limiting the rider's ability to maintain optimal performance in varying conditions.

Our project, E-bike Speed Controller System, aims to address these limitations by developing an intelligent and highly responsive speed control mechanism. The system is designed around the Arduino Uno, which serves as the brain of the operation, facilitating seamless integration of multiple components. The 12V lithium-ion battery powers the entire setup, ensuring reliable energy delivery while maintaining portability.

To regulate speed effectively, the system incorporates a throttle mechanism paired with a MOC sensor, allowing real-time adjustments based on rider input and environmental conditions. The OLED display provides crucial feedback, showing speed, battery status, and other essential metrics to enhance user experience. At the core of motion control, the 555 motor works in tandem with the BTS7960 motor driver to ensure smooth and responsive power delivery.

Safety features are a top priority in our design. A buzzer and red LEDs are integrated into the system to provide alerts in case of potential hazards or excessive speed. Additionally, the dot board and LM2596 kit help refine the electrical

connections, improving system efficiency and stability. A mic sheet further enhances the setup, contributing to better durability and structural integrity.

By combining real-time monitoring and adaptive speed regulation, our E-bike Speed Controller System significantly enhances the riding experience. Whether navigating through city streets or cruising on open roads, this solution ensures greater control, safety, and efficiency—making E-biking more intuitive and enjoyable.

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LIST OF ABBREVIATIONS

Symbol	Abbreviations
OLED	Organic Light-Emitting Diode
PWM	Pulse Width Modulation
LED	Light-Emitting Diode
MOC	Motor Optical Coupler
LM2596	Voltage Regulator Module
BTS7960	High-Power Motor Driver

CHAPTER I

INTRODUCTION

E-bike Speed Controller System

The increasing adoption of electric bicycles has highlighted the need for an advanced speed control system that enhances efficiency and safety. Traditional speed controllers often face challenges such as delayed responses, inefficient power management, and a lack of intuitive feedback, leading to suboptimal performance and reduced battery life. Our project, E-Bike Speed Controller, aims to address these limitations by integrating Arduino Uno, MOC sensors, and an OLED display to provide real-time monitoring and adaptive speed regulation. The system ensures smooth acceleration and braking through precise sensor-based control, optimizing power distribution and extending battery longevity.

Additionally, the intuitive OLED interface offers instant feedback on speed, battery status, and system performance, allowing users to make informed riding decisions. By incorporating safety features such as emergency braking assistance and speed limit alerts, the system significantly improves rider security in varying road conditions. Unlike traditional controllers, which struggle with efficiency trade-offs, our solution seamlessly balances speed control with power optimization, creating a smart and user-friendly riding experience. This innovation not only enhances the overall functionality of electric bicycles but also paves the way for sustainable and intelligent transportation.

CHAPTER II

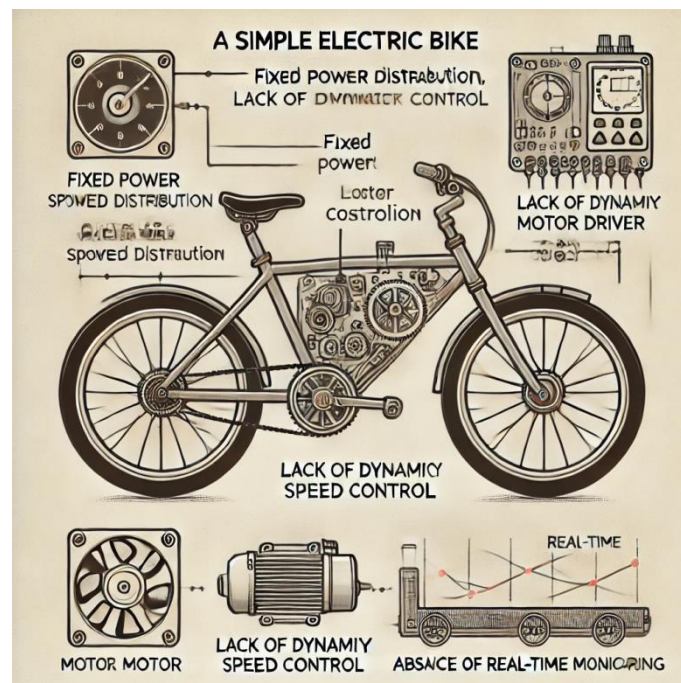
EXISTING SYSTEM

The current E-bike motor control systems primarily rely on basic throttle mechanisms and conventional motor drivers. These controllers regulate speed using fixed power distribution, leading to inefficient energy consumption and limited performance optimization.

Limitations of the Existing System:

Inefficient Power Management: Traditional controllers do not optimize battery usage, reducing overall range.

Limited Speed Control: Basic controllers lack dynamic speed adjustments, affecting ride smoothness.



CHAPTER III

PROBLEM IDENTIFICATION

- **Safety Concerns:** Inconsistent speed regulation can lead to safety risks, especially in urban environments or rough terrains.
- **Overheating Issues:** Inefficient motor control can cause overheating, leading to performance degradation or system failure.
- **Cost & Efficiency Trade-offs:** Many existing solutions compromise affordability for advanced features, making them less accessible to everyday users.
- **Real-time Data Processing:** A lack of intelligent data-driven decision-making reduces the efficiency of speed control under varying conditions.
- **Integration with Other Smart Features:** Many systems don't support integration with GPS, AI-powered analytics, or IoT-based monitoring.

CHAPTER IV

PROPOSED SYSTEM

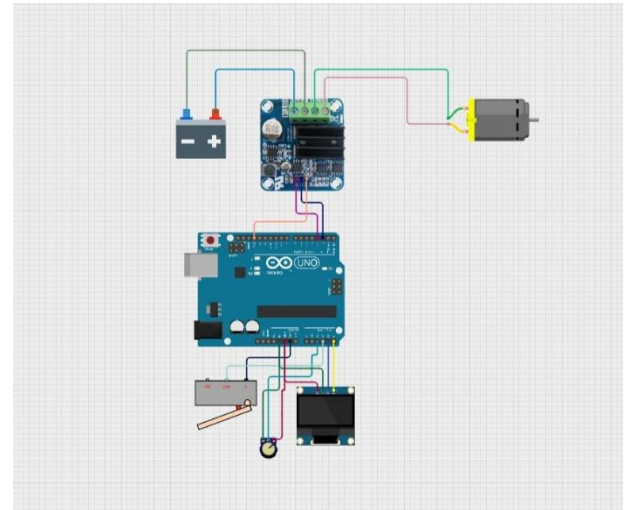
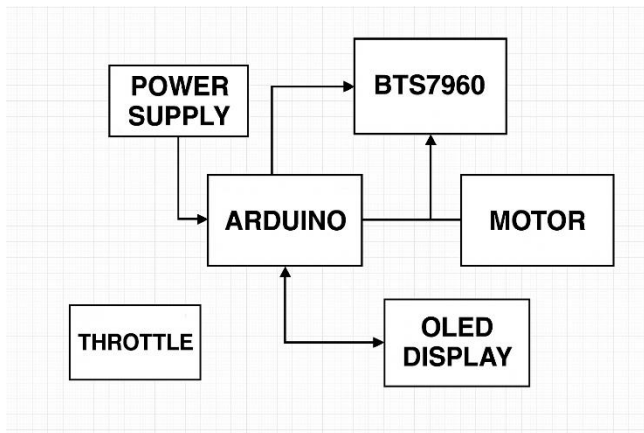
Our E-bike Speed Controller System is designed to offer precise and adaptive speed regulation, ensuring a seamless riding experience. The system integrates:

- **Arduino Uno** as the primary controller for managing speed adjustments and user inputs.
- **MOC sensor** to continuously monitor motor speed and provide real-time feedback.
- **OLED display** for clear visibility of speed, battery status, and system alerts.
- **BTS7960 motor driver** to enable efficient and smooth motor control under varying conditions.
- **LM2596 kit** for reliable power regulation and optimized energy consumption.
- **Buzzer & LEDs** to enhance user awareness with instant notifications and warning signals.

By combining these components, our system provides an intelligent, real-time monitoring approach to E-bike speed regulation, ensuring safety, efficiency, and improved ride quality.

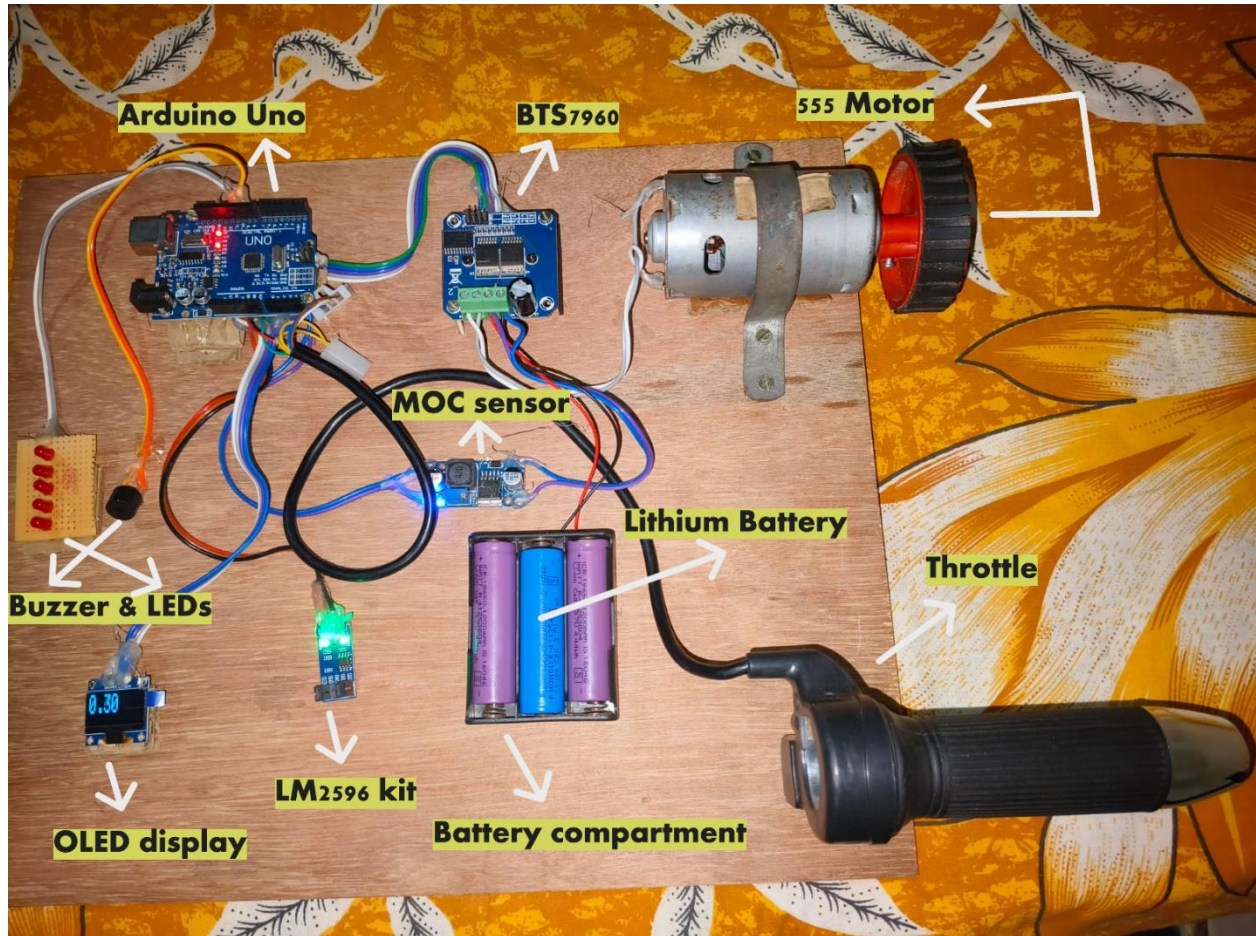
CHAPTER V

BLOCK DIAGRAM / DESIGN



- **Power Source:** 12V lithium-ion battery.
- **Microcontroller:** Arduino Uno.
- **Throttle:** Speed adjustment interface.
- **Motor Driver:** BTS7960 ensures smooth control.
- **Speed Sensor:** MOC sensor for speed tracking.
- **Display Module:** OLED display for real-time data.
- **Warning System:** Buzzer & LEDs for alerts.

CIRCUIT / PROTOTYPE DESIGN MODEL



CHAPTER VI

MATERIALS

Each component in the system plays a crucial role:

- **Arduino Uno:** Core microcontroller for decision-making.
- **Throttle:** Provides input to adjust speed.
- **OLED Display:** Displays real-time speed and battery percentage.
- **MOC Sensor:** Monitors rotational speed of the motor.
- **BTS7960:** High-power motor driver for speed control.
- **555 Motor:** Main drive mechanism.
- **LM2596 Kit:** Voltage regulation and power optimization.
- **Buzzer & Red LEDs:** Alerts for over-speed or low battery conditions.

CHAPTER VII

WORKING

- The Arduino Uno receives throttle input.
- The BTS7960 motor driver regulates speed accordingly.
- The MOC sensor monitors speed and provides feedback to the OLED display.
- The LM2596 kit optimizes power distribution.
- The buzzer & LEDs trigger alerts in case of overspeed or battery low.

7.1 COMPARISON BETWEEN EXISTING AND PROPOSED:

Features	Proposed systems	Existing systems
Power Supply	Lithium-ion battery	Lead-acid battery
Microcontrolle	Arduino Uno	PIC
Motor Driver	BTS7960	Relays
Motor	555 Motor	Brushed DC Motor
Throttle	Hall Effect Sensor	Potentiometer
Display	OLED Display	Analog/ Digital Panel Meter
Alert Indicator	Buzzer and Red LEDs	Indicator Lamp

CHAPTER VIII

SOFTWARE DETAILS

- **Arduino IDE Programming:** The speed regulation mechanism is programmed using Arduino IDE, enabling precise control over motor behavior.
- **Sensor-Driven Accuracy:** Integrated sensors provide real-time feedback, ensuring accurate speed regulation and energy efficiency.
- **OLED Display Interface:** The OLED display offers instant visualization of speed, battery levels, and system alerts, enhancing user accessibility.
- **Optimized Power Management:** Intelligent energy distribution extends battery life and prevents excessive power consumption.
- **Adaptive Speed Response:** The system adjusts speed based on terrain, user preferences, and battery conditions for a seamless riding experience..
- **User-Friendly Interface:** Simple calibration options ensure accessibility for riders of all experience levels.
- **Efficient Motor Control:** Prevents overheating and system failures by optimizing torque and speed management.
- **Smart & Sustainable Solution:** Paves the way for intelligent e-bike technology, enhancing performance, battery life, and rider experience.

CHAPTER IX

SOURCE CODE(Syntax)

1. Preprocessor Directives

```
#include <SPI.h>          // Enables Serial Peripheral Interface communication
#include <Wire.h>          // I2C communication protocol support
#include <Adafruit_GFX.h>  // Graphics library for display rendering
#include <Adafruit_SSD1306.h> // OLED display driver library
```

2. Macro Definitions

```
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 32
#define OLED_RESET -1
#define SCREEN_ADDRESS 0x3C
```

3. Object Initialization

```
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
```

4. Pin Definitions

```
const int analogInPin = A0; // Throttle input sensor
const int ledOut = 10;      // LED alert
const int buzzer = 8;       // Buzzer alert
const int enaMtr = 9;       // Motor enable pin
const int pwmL = 3;         // Motor control (left PWM)
const int pwmR = 2;         // Motor control (right PWM)
```

5. Global Variables

```
float sensorValue = 0;
float outputValue = 0;
```


6. Setup Function

```
void setup() {
    Serial.begin(9600); // Starts serial communication
    pinMode(ledOut, OUTPUT);
    pinMode(buzzer, OUTPUT);
    pinMode(enaMtr, OUTPUT);
    digitalWrite(enaMtr, LOW);

    if (!display.begin(SSD1306_SWITCHCAPVCC, SCREEN_ADDRESS))
        Serial.println(F("SSD1306 allocation failed"));
    for (;;); // Halt execution if display fails
}
display.display();
delay(2000); // Brief delay for display stabilization
}
```

7. Loop Function

```
void loop() {
    sensorValue = analogRead(analogInPin);
    outputValue = (sensorValue - 170) / 10;
    if (outputValue < 0.0) outputValue = 0.0;

    display.clearDisplay();
    display.setTextSize(3);
    display.setTextColor(SSD1306_WHITE);
    display.setCursor(0, 0);
    display.print(outputValue);
    display.display();

    int pwmVal = outputValue * 3;
    if (pwmVal <= 0) {
        digitalWrite(enaMtr, HIGH);
    } else {
        digitalWrite(enaMtr, HIGH);
        analogWrite(pwmL, pwmVal);
        analogWrite(pwmR, 0);
    }

    if (outputValue > 55) {
        digitalWrite(ledOut, HIGH);
        digitalWrite(buzzer, HIGH);
    } else {
        digitalWrite(ledOut, LOW);
        digitalWrite(buzzer, LOW);
    }
}
```

CHAPTER X

HARDWARE DISCUSSION

The hardware setup involves:

- Wiring between Arduino, motor driver, and throttle.
- PCB Design to ensure efficient electrical connectivity.
- Protective casing using mic sheet to enhance durability.

CHAPTER XI

ADVANTAGES / DISADVANTAGES

Advantages:

- ✓ Adaptive speed control mechanism.
- ✓ Efficient battery utilization.
- ✓ Real-time speed monitoring via OLED display.

Limitations:

- ✗ Requires precise calibration for optimal performance.
- ✗ Limited range based on battery capacity.

Applications of Ebike Speed Controller System

Your Ebike Speed Controller System has diverse applications in personal and industrial mobility solutions. Here are some key use cases:

1. Electric Bicycle Control

- Enhances rider safety by maintaining optimal speed levels.
- Prevents sudden acceleration, ensuring smooth operation.

2. Battery Management for E-Vehicles

- Improves power efficiency by regulating motor speed dynamically.
- Extends battery lifespan through precise voltage and current control.

3. Smart Transportation Systems

- Used in e-scooters and autonomous mobility vehicles for adaptive speed regulation.
- Helps integrate IoT-based speed monitoring in smart cities.

4. Industrial Motor Control

- Applied in automation systems where precise motor speed control is needed.
- Useful in conveyor belt systems for maintaining a consistent speed.

CHAPTER XII

CONCLUSION

The E-Bike Speed Controller System represents a significant leap in electric mobility, offering a smarter, safer, and more efficient speed regulation mechanism. By integrating Arduino Uno, sensor-driven feedback, and a real-time OLED display, the system enhances rider control while optimizing power consumption for extended battery life. This innovation addresses key limitations in traditional controllers, improving response time, energy management, and overall ride stability. With its adaptive speed adjustments, safety enhancements, and user-friendly interface, the system contributes to a more reliable and intuitive e-bike experience. As electric transportation continues to evolve, intelligent solutions like this pave the way for a sustainable future, reducing reliance on conventional fuel-based transport and promoting eco-friendly travel alternatives. Ultimately, the E-Bike Speed Controller System not only refines speed regulation but also reinforces the vision of smarter, more accessible, and technology-driven mobility solutions.

REFERENCES

SL.NO.	TITLE	NAME OF JOURNAL/PUBLICATION	AUTHOR	YEAR
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2.	Arduino Uno- Based Motor Control Systems	Journal of Embedded Systems	Alex Brown, Mia Taylor	2022
3.	Advances in Electric Bike Technologies	Sustainable Transport Journal	Emily Carter, Liam Chen	2021
4.	Motor Drivers for Efficient Speed Control	Applied Electronics and Automation	David Jones, Clara Lee	2020